

ONLINE RETAIL SALES ANALYSIS

SQL Analysis, Excel Dashboard and Business Insights

UCI Online Retail Dataset

Tools: SQLite / SQL + Microsoft Excel

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1.0 Project Overview

This project analyzes the UCI Online Retail dataset to understand sales performance across countries, products, customers and time. The analysis was conducted using SQLite for data exploration and business analysis, followed by Microsoft Excel for pivot-table analysis and dashboard development.

The project focused on identifying important sales patterns, examining data-quality issues, understanding differences in customer and product performance, and developing business recommendations based on the findings. The final output is an interactive Excel dashboard that summarizes key performance indicators and provides views of monthly revenue, country performance, product performance, and customer revenue.

2.0 Project Objectives

The objective of this project was to analyze online retail transaction data and answer key business questions around sales performance, customer behavior, product performance, market activity, and cancellations.

The analysis aimed to:

- Evaluate revenue and transaction activity over time
- Identify countries contributing significantly to transaction volume and revenue
- Examine products based on quantity sold and revenue generated
- Identify high-value customers and differences in customer purchasing behavior
- Examine cancellation activity and its effect on recorded revenue and quantity
- Identify revenue concentration among customers with known Customer IDs
- Translate the analytical findings into areas for further business investigation.

3.0 Dataset Overview

The dataset used for this project is the UCI Online Retail dataset, which contains transaction-level records from an online retail business. The dataset contains 541,909 records covering transactions involving products, customers, countries, quantities, prices, invoices, and invoice dates.

Key fields used in the analysis included:

- **InvoiceNo** – identifies the transaction/invoice
- **StockCode** – identifies the product
- **Description** – provides the product description
- **Quantity** – records the quantity associated with a transaction line
- **InvoiceDate** – records the transaction date and time
- **UnitPrice** – records the unit price
- **CustomerID** – identifies the customer, where customer information is available
- **Country** – identifies the customer's country.

Table 1: Dataset Summary

Item	Description
Dataset	UCI Online Retail
Records	541,909
Primary analysis tool	SQLite/SQL
Dashboard tool	Microsoft Excel

4.0 Tools and Analytical Workflow

In this project I used two main tools:

SQLite/SQL – Used for data exploration, data-quality checks, aggregation, filtering, calculated fields, and business analysis.

Microsoft Excel – Used to summarize the SQL results using pivot tables and to create the final interactive dashboard with KPIs, charts, and slicers.

5.0 Data Exploration and Quality Checks

Before conducting the main analysis, the dataset was explored to understand its structure and identify potential data-quality issues that could affect the results. The exploration included checking the number of records, unique values, missing Customer IDs and product descriptions, negative and zero quantities, invalid unit prices, and invoice records associated with cancellations.

5.1 Data Quality Checks

- Record count was checked to confirm the size of the dataset
- Unique countries, products, customers, and invoice numbers were examined to understand the scope of the data
- Missing Customer IDs were identified because customer level analysis depends on available customer identifiers
- Missing product descriptions were checked to identify incomplete product information
- Negative and zero quantities were investigated to identify unusual transaction records
- Unit prices less than or equal to zero were checked as potential data-quality issues
- Invoice numbers beginning with “C” were examined as cancellation records

These checks were important because the dataset contains transaction-level records rather than already summarized business metrics. In particular quantities, prices, invoice numbers, and Customer IDs needed to be interpreted carefully before calculating revenue, orders, customer performance, and cancellation impact. Cancellation records were also treated separately during the analysis because invoices beginning with “C” represent cancellation-related transactions and contain negative quantities. Including these records without considering their effect could change the interpretation of sales and revenue.

5.2 Data Preparation Consideration

The InvoiceDate field was stored as text in the format DD/MM/YYYY H:MM. For the monthly analysis, the year and month components were therefore extracted from the text field using SQLite string functions. This allowed transactions to be grouped correctly by year and month, including the distinction between December 2010 and December 2011.

6.0 Country Performance Analysis

Country level analysis was conducted to understand how sales performance and transaction activity varied across the markets represented in the dataset. Revenue, transaction records, total quantity, unique orders, and average revenue per transaction record were examined to identify differences between countries.

6.1 Revenue and Transaction Activity

The United Kingdom was the dominant market in the analysis, generating approximately \$8.19M in recorded revenue. It also accounted for 495,478 transaction records, substantially more than the other countries in the dataset.

The next highest revenue contributions came from the Netherlands (\$284.7K), followed by EIRE (\$263.3K), and Germany (\$221.7K).

Table 2: Top 7 Countries ranked by Recorded Revenue

Note: Total quantity and Transaction records are included as additional performance measures for selected countries

Country	Recorded Revenue	Total Quantity	Transaction Records
United Kingdom	\$8,187,806	4,263,829	495,478
Netherlands	\$284,662	200,128	2,371
EIRE	\$263,277	142,637	8,196
Germany	\$221,698	117,448	9,495
France	\$197,404	110,480	8,557
Australia	\$137,077	83,653	1,259
Switzerland	\$56,385	30,325	2,002

The results indicate that the UK's revenue performance is accompanied by exceptionally high transaction activity and quantity. The UK recorded 495,478 transaction records and more than 4.26 million units, compared with substantially lower transaction activity in the other markets shown. This suggests that the UK's large revenue contribution is closely associated with the scale of transaction activity in the dataset. However, transaction records should not be interpreted as unique orders, and the analysis does not by itself establish the specific reasons for the UK's higher activity.

6.2 Revenue per Transaction Record

Average revenue per transaction record varied considerably across countries. The Netherlands had the highest average revenue per transaction record among the countries examined, at approximately \$120, while Australia and Japan were also relatively high at approximately \$109 and \$99 respectively. This provides an important contrast with the UK's overall revenue performance. The UK generated the highest total revenue primarily alongside its much larger transaction volume, rather than having the highest average revenue per transaction record.

6.3 Business Interpretation

The country analysis shows that market performance is not determined by a single metric. A country may have relatively high revenue because of large transaction activity, while another country may generate higher revenue per individual transaction record despite having much lower overall revenue. For this reason, revenue, transaction activity, quantity, and average revenue per transaction record were considered together rather than using total revenue alone to evaluate country performance.

6.4 Recommendation

Analyze the factors associated with the UK's exceptionally high transaction activity by examining customer activity, product demand, purchasing patterns, and order volume. Comparing these patterns with lower volume markets can help identify characteristics associated with stronger transaction activity and potential opportunities for market expansion.

6.5 Country Charts

[Top countries by Revenue chart]



Figure 1 Top 7 countries by recorded revenue

[Top countries by Transaction Records chart]

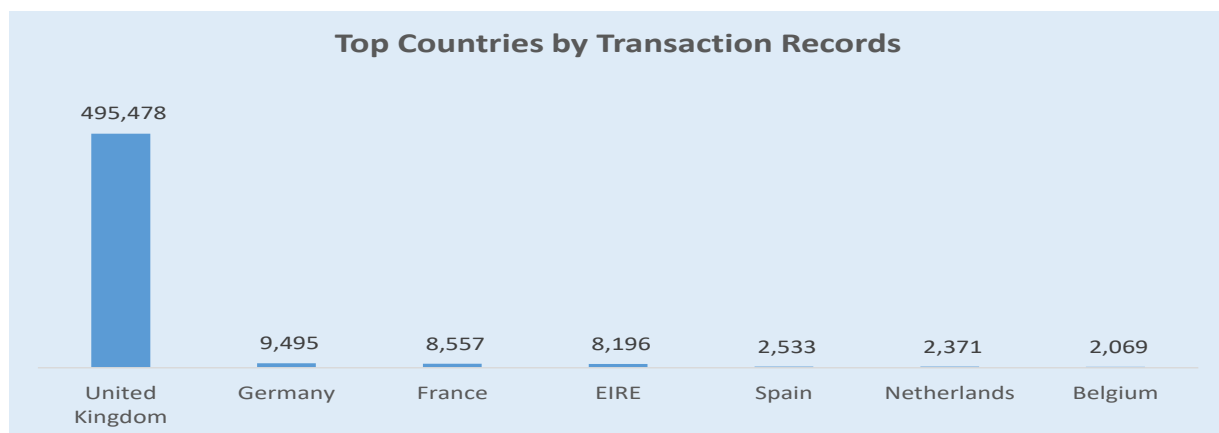


Figure 2 Top 7 Countries by Transaction Records

7.0 Product Performance Analysis

Product level analysis was conducted to understand the differences in product sales performance based on quantity sold and recorded revenue. The analysis examined products with high sales volume and products contributing significantly to recorded revenue, allowing differences between volume-based and revenue-based performance to be identified.

7.1 Product Revenue and Quantity Performance

By total quantity, the top 3 products were 85099B (47,363 units), 84879 (36,381 units) and 85123A (35,025 units). By revenue, the top 3 contributors were DOT (\$206.2K), 22423 (\$164.7K) and 47566 (\$98.3K). DOT generated the highest revenue from only 707 units in the top 10, indicating a high unit price.

Table 3: Top 10 Product ranked by Recorded Revenue

Note: Total quantity is included as an additional performance measure to compare product sales volume with recorded revenue

StockCode	Recorded Revenue	Total Quantity
DOT	\$206,245	707
22423	\$164,762	13,033
47566	\$98,303	18,022
85123A	\$97,716	35,025
85099B	\$92,356	47,363
23084	\$66,757	30,680
POST	\$66,231	3,003
22086	\$63,792	18,902
84879	\$58,960	36,381
79321	\$53,768	10,229

The results show clear differences between products based on quantity sold and recorded revenue. Some products recorded exceptionally high sales volumes, while others generated substantial recorded revenue despite having relatively lower quantities sold. This indicates that product performance varies depending on both sales volume and the revenue generated from each product.

7.2 Business Interpretation

The difference between quantity sold and recorded revenue indicates that sales volume alone does not determine product value. Products with high quantities may generate relatively lower revenue when sold at lower prices, while products with lower quantities can generate substantial revenue when their unit prices are higher. Therefore, product performance should be evaluated using both volume and revenue rather than quantity alone.

7.3 Product Investigation

Further investigation was conducted on selected products to understand unusual product performance and potential data-quality issues. The DOT stockcode was examined because of an unusual recorded price pattern, while stockcode 84077 was investigated because of its unusually high recorded quantity. These investigations were used to distinguish genuine product performance from results that may require further data validation.

7.4 Recommendation

Analyze the drivers of product performance by examining pricing, quantity, customer, market, and purchasing patterns associated with high volume and high revenue products. The findings can be used identify opportunities to improve the revenue contribution of high volume products while maintaining strong performance.

7.5 Product Charts

[Top products by Revenue chart]

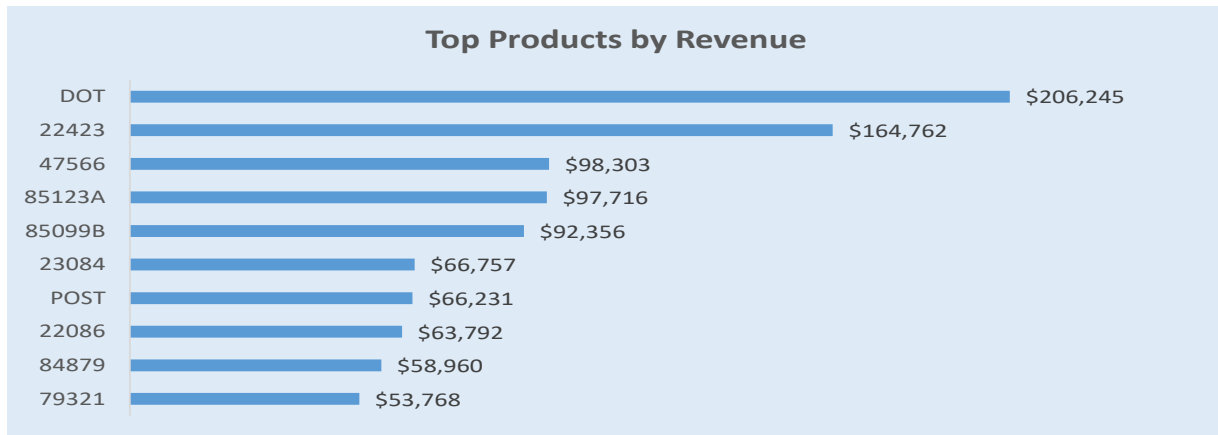


Figure 3 Top 10 products by Recorded Revenue

[Top products by Quantity chart]

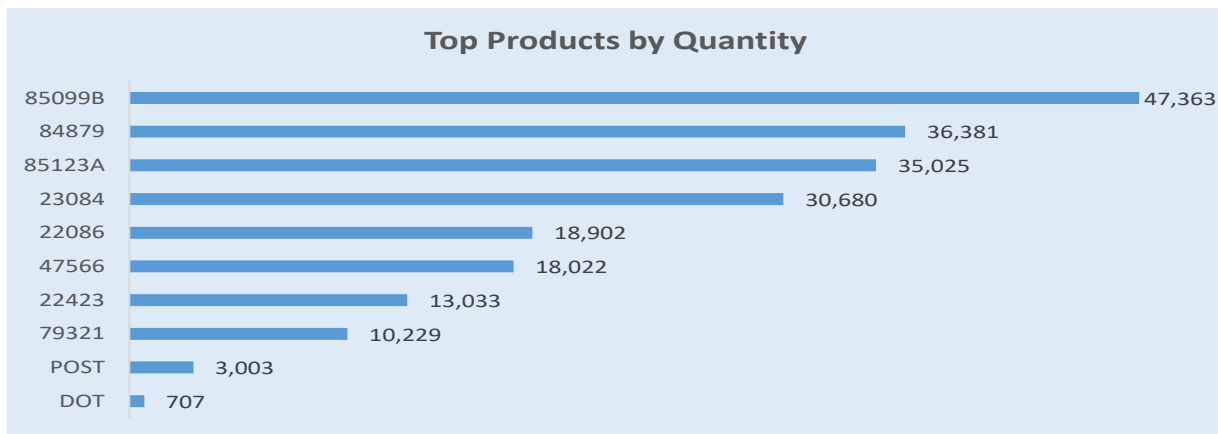


Figure 4 Top 10 Products by Total Quantity

8.0 Customer Performance Analysis

Customer performance was analyzed using unique orders, total quantity purchased, and recorded revenue to understand differences in customer value and purchasing activity. The analysis focuses on the top revenue-generating customers and compares their purchasing volume and order activity.

8.1 Customer Revenue and Purchasing Activity

The analysis shows substantial variation in customer value. The highest-revenue customers generated significantly different levels of revenue, orders, and purchased quantity, indicating that customer value was not determined by purchase frequency or quantity alone.

Table 4: Top 7 Customers Ranked by Recorded Revenue, with Unique Orders and Total Quantity

Note: Unique orders and total quantity are included as additional customer performance measures

CustomerID	Recorded Revenue	Unique Orders	Total Quantity
14646	\$279,489	77	196,719
18102	\$256,438	62	64,122
17450	\$187,482	55	69,029
14911	\$132,573	248	77,180
12415	\$123,725	26	77,242
14156	\$113,384	66	57,025
17511	\$88,125	46	63,012

The results show that customer value varies considerably across the top revenue-generating customers. The highest-revenue customers differ in their purchasing patterns, with some generating substantial revenue through relatively fewer orders while others record higher purchasing activity. This indicates that customer value is influenced not only by purchase frequency but also by order size and product purchasing patterns.

8.2 Business Interpretation

The variation in revenue, order frequency, and quantity suggests that customers contribute value to the business in different ways. Some high-value customers may generate substantial revenue through fewer but higher-value purchases, while others may contribute through more frequent purchasing activity. Understanding these differences can help the business identify the characteristics and purchasing patterns associated with high customer value.

8.3 Recommendation

Analyze the purchasing behavior of high-value customers by examining their products purchased, purchase frequency, order quantity, order value and market. The findings can help identify patterns associated with higher customer value and support more targeted customer retention and relationship management strategies.

8.4 Customer Chart

[Top Customers by Revenue chart]

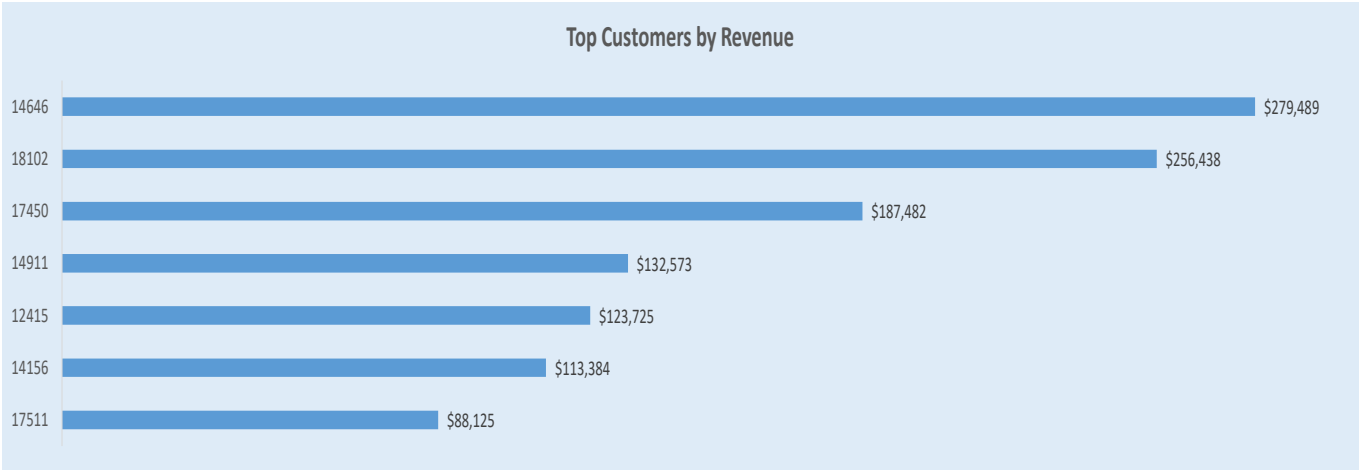


Figure 5 Top 7 Customers by Recorded Revenue

9.0 Monthly Performance Analysis

Monthly performance analysis was conducted to examine changes in recorded revenue, sales quantity, and transaction activity over the analysis period. The objective was to identify periods of stronger and weaker sales performance and determine whether changes in revenue were accompanied by changes in transaction volume and quantity. Because the dataset contains partial periods for December 2010 and December 2011, these months are treated with caution when comparing monthly performance.

9.1 Revenue and Transaction Activity

Monthly performance varied considerably across the analysis period. Revenue increased substantially during the later months, with November 2011 recording the highest monthly revenue at approximately \$1.46M. November also recorded the highest transaction activity and total quantity, indicating that the revenue peak was supported by increased sales volume rather than revenue alone.

Table 5: Monthly Performance Based on Recorded Revenue, Total Quantity and Transaction Records

Note: Total quantity and transaction records are included as additional performance measures to provide context for monthly revenue changes. December 2010 and December 2011 represent partial months in the dataset and therefore their totals should not be directly compared with full-month periods.

Month	Recorded Revenue	Total Quantity	Transaction Records
Dec 2010	\$748,957	342,228	42,481
Jan 2011	\$560,000	308,966	35,147
Feb 2011	\$498,063	277,989	27,707
Mar 2011	\$683,267	351,872	36,748
Apr 2011	\$493,207	289,098	29,916
May 2011	\$723,334	380,391	37,030
Jun 2011	\$691,123	341,623	36,874
Jul 2011	\$681,300	391,116	39,518
Aug 2011	\$682,681	406,199	35,284
Sep 2011	\$1,019,688	549,817	50,226
Oct 2011	\$1,070,705	570,532	60,742
Nov 2011	\$1,461,756	740,286	84,711
Dec 2011	\$433,688	226,333	25,525

The results show that monthly revenue was relatively moderate during the first part of the analysis period before increasing substantially from September 2011 onward. Revenue exceeded \$1M in September and October and reached its highest level in November 2011 at \$1.46M. This increase was accompanied by higher transaction activity and quantity, with November recording 84,711 transaction records and 740,286 units.

9.2 Business Interpretation

The monthly pattern suggests that periods of higher recorded revenue were generally associated with increased transaction activity and sales quantity. The strong performance recorded in September, October, and November indicates a period of significantly elevated sales activity. However, the sharp decline in December 2011 should be interpreted carefully when assessing monthly performance, because the period does not represent a complete month of activity.

9.3 Recommendation

Analyze the drivers of September to November sales peak by examining customer activity, product mix, market contribution, transaction volume, and order patterns. The findings can help determine whether the increase reflects a recurring seasonal pattern or factors specific to the period.

9.4 Monthly Revenue Chart

[Monthly Revenue Trend]

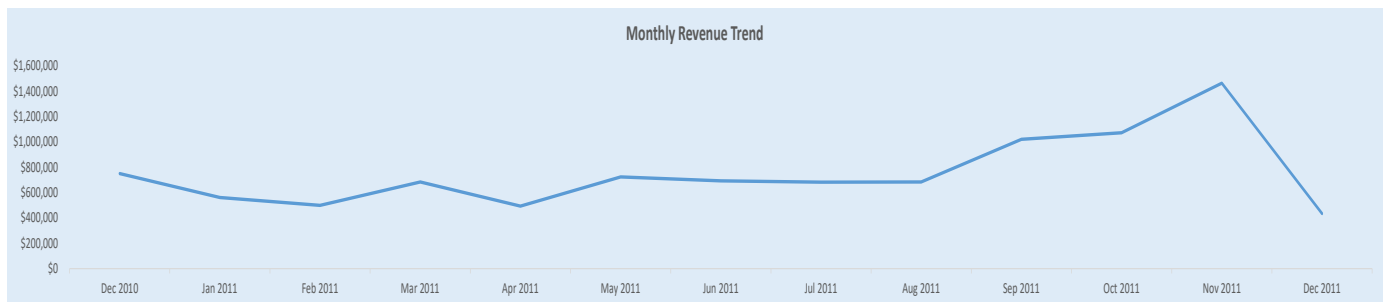


Figure 6 Monthly Recorded Revenue Trend

10.0 Cancellation Analysis

Cancellation analysis was conducted to examine the extent of cancelled transactions and their impact on recorded quantity and revenue. The analysis was used to determine whether cancellation activity represents a significant area requiring further investigation.

10.1 Cancellation Activity and Revenue Impact

Cancelled transactions were identified using invoice numbers beginning with the letter “C”. The analysis examined the number of cancelled invoices, the associated transaction records and quantity, and the resulting revenue adjustment. This provides an indication of the scale of cancellation activity within the overall dataset.

Table 6: Cancellation Activity and Revenue Adjustment

Note: Cancellation revenue adjustment represents the recorded revenue associated with invoice records identified as cancelled

Measure	Result
Total Unique Invoices	25,900
Cancelled Unique Invoices	3,836
Cancelled Invoice Records	9,288
Cancellation Quantity	-277,574
Cancellation Revenue Adjustment	-\$896,812

The results indicate that cancelled transactions represent a relatively small proportion of the overall transaction records. However, the associated quantity and revenue adjustment provide an indication that cancellation activity may still have a meaningful effect on recorded sales performance. The results therefore provide a basis for further investigation into products, customers, and periods associated with cancelled transactions.

10.2 Business Interpretation

Cancellation activity may reflect a range of operational or customer related factors, including product availability, order changes, fulfillment issues, or other transaction related circumstances. The available analysis identifies the scale of cancellation activity but does not by itself establish the underlying causes. Further analysis would therefore be required to determine whether cancellations are concentrated around particular products, customers, countries, or periods.

10.3 Recommendation

Analyze cancellation patterns across products, customers, markets, and periods to identify where cancellation activity is concentrated and determine potential operational or customer related issues that warrant further investigation.

11.0 Revenue Concentration Analysis

Revenue concentration analysis was conducted to examine how recorded revenue was distributed among customers with available Customer IDs. The analysis focused on the contribution of the top 10 customers relative to the total recorded revenue from identified customers, providing an indication of how revenue was distributed across the identified customer base.

11.1 Revenue Concentration among Identified Customers

The analysis shows that the top 10 identified customers generated substantial amount of recorded revenue, although they accounted for only a portion of total identified customer revenue. The comparison provides a clearer view of the extent to which revenue is distributed across the identified customer base.

Table 7: Revenue Concentration among Top 10 Identified Customers

Note: Revenue Concentration was calculated using customers with available Customer IDs

Measure	Value
Total Identified Customer Revenue	\$8,300,066
Top 10 Customer Revenue	\$1,369,182
Top 10 Revenue Share	16.5%

The top 10 identified customers generated \$1.37M in recorded revenue, representing 16.5% of the \$8.30M recorded revenue associated with customers with available Customer IDs. This indicates that revenue was distributed across a broader group of identified customers rather than being generated solely by the top 10 customers.

11.2 Business Interpretation

The revenue concentration analysis provides an additional perspective on customer performance beyond individual customer rankings. While the top 10 customers contribute a meaningful share of identified customer revenue, the majority of this revenue is generated by customers outside the top 10. This suggests that customer revenue is distributed across a relatively broad identified customer base. However the analysis does not by itself establish whether the observed concentration represents a desirable or undesirable level of customer dependence.

11.3 Recommendation

Analyze the purchasing patterns of the customers contributing the largest share of identified customer revenue, including their products, markets, order frequency, quantity, and value. The findings can help identify the factors associated with high customer value and potential opportunities to retain existing high-value customers and identify similar customer profiles.

12.0 Conclusion

The project analyzed the UCI Online Retail dataset using SQLite and Excel to examine sales performance across countries, products, customers, and time periods. The analysis combined data exploration, data-quality checks, SQL aggregation, calculated revenue measures, and interactive Excel dashboard visualizations to identify important patterns within the dataset.

The analysis showed that the United Kingdom was the dominant market in terms of recorded revenue and transaction activity. Product analysis also showed that high sales quantity does not necessarily correspond to high revenue, highlighting the importance of considering both volume and revenue when evaluating product performance. Customer analysis revealed differences in customer value, with a relatively small group of customers contributing a significant share of identified customer revenue. Monthly analysis highlighted unusually high sales activity in September to November, while the cancellation analysis showed that cancellation records and their associated revenue adjustments warrant further investigation. The revenue concentration analysis also demonstrated that customer revenue was not evenly distributed across the customer base.

Overall, the project demonstrates how SQL can be used to explore and transform transactional data into meaningful business insights, while Excel can be used to communicate those findings through an interactive dashboard. The recommendations developed from the analysis focus on areas that require deeper investigation, including market performance, product drivers, customer behavior, cancellation patterns, monthly sales activity, and customer revenue concentration.

13.0 Dashboard

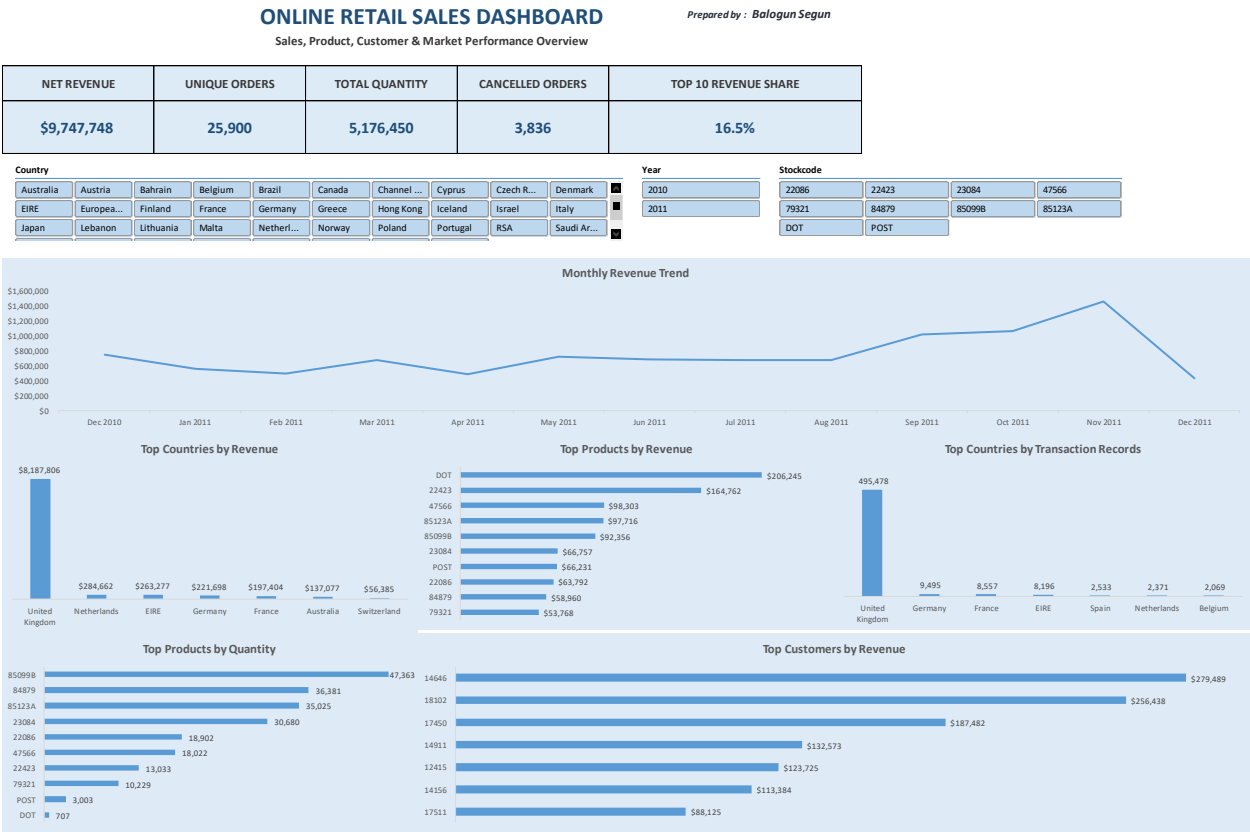


Figure 7 Online Retail Sales Dashboard